

01 | Day 1: See It

How can simple operations produce intelligent-looking behavior?

Retrieve: In one sentence, what do you currently think a neural network does?

02 | Today's Investigation

Choose it -> Move it -> Break its limit -> Transform it -> Trace it

Morning	Afternoon
model choice	layers and shapes
one-neuron geometry	activation behavior
linear success and XOR	forward network mystery

Outcome: Explain one correct and one incorrect prediction from evidence.

03 | Neural Networks Are One Toolbox

AI -> machine learning -> deep learning -> some generative AI

AI -> rules / search / planning

Machine learning -> linear / tree / other models

Deep learning -> classify / forecast / detect

Prompt: Where would a simple baseline live?

04 | Choose the First Experiment

Case A: 1,800 labeled rows, 10 named numeric features, transparent result needed tomorrow.

Case B: 50,000 labeled product images under varied lighting.

For each:

1. Choose the simplest credible first model family.
2. Name evidence that would make you change course.

05 | Training Is Not Inference

Training: examples + targets -> changed parameters

Inference: new inputs + fixed parameters -> outputs

Today: fixed parameters, forward evidence.

Tomorrow: objective, sensitivity, updates.

06 | One Neuron, Three Stages

$$z = w \cdot x + b \longrightarrow p = \sigma(z) \longrightarrow \hat{y} = \mathbb{1}[p \geq 0.5]$$

Object	Meaning
w	evidence multipliers
b	adjustable offset
z	raw score / logit
p	probability-shaped output

Boundary: $w \cdot x + b = 0$

07 | ACT-D1-01: Predict Boundary Motion

Before any slider moves, sketch three cases:

1. increase `w1` only;
2. increase `w2` only;
3. increase `b` only.

For each, predict:

orientation | location | one probe probability

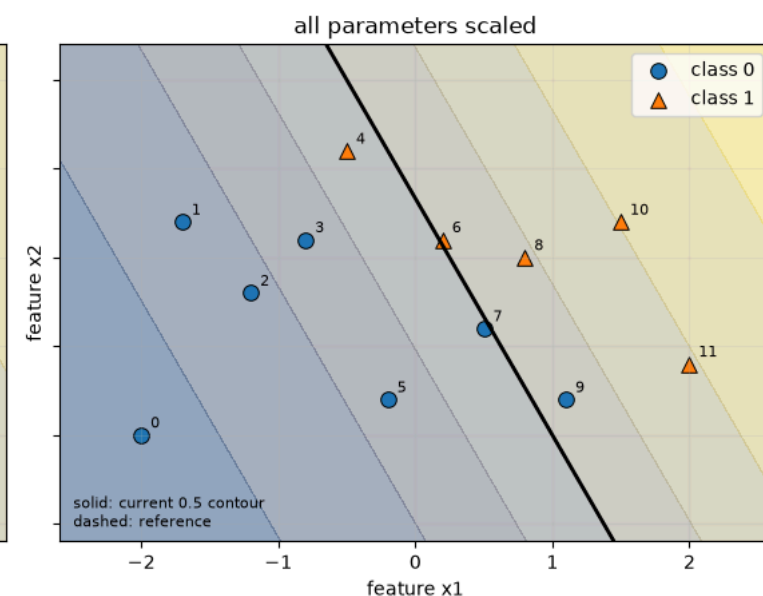
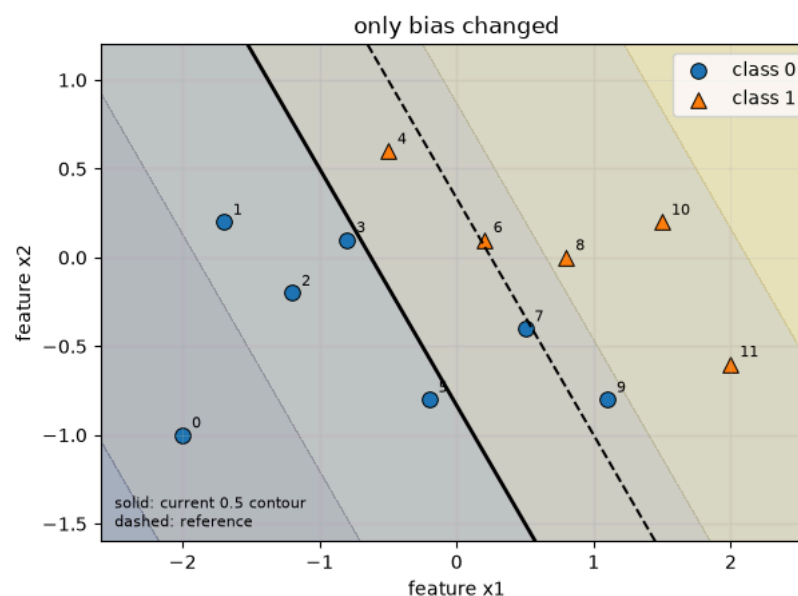
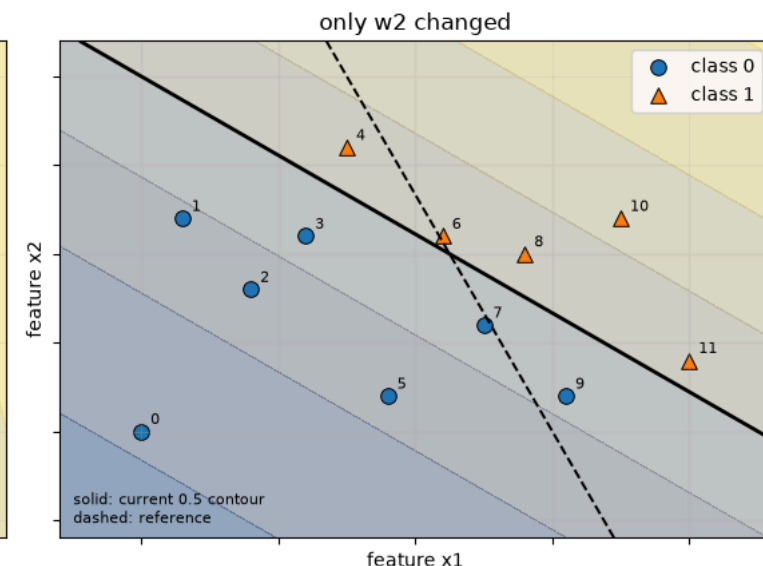
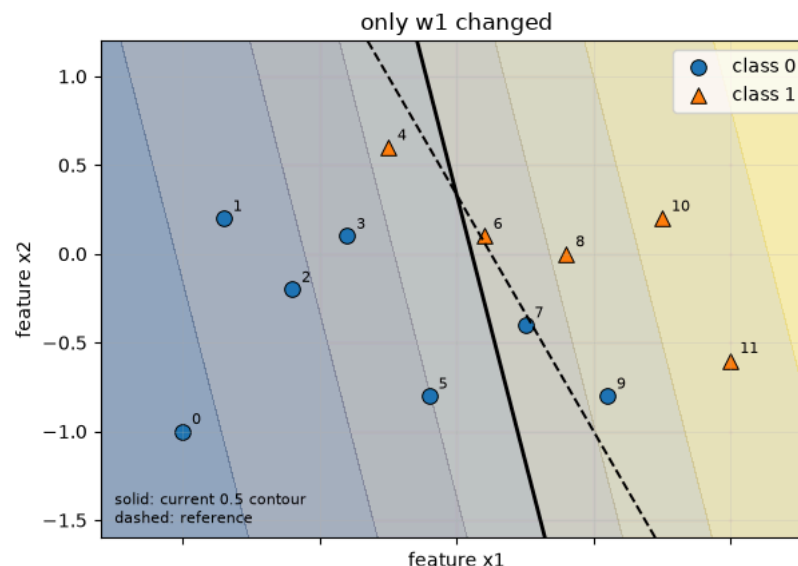
Mark the claim you trust least.

08 | Boundary Motion Evidence

Ghost = before | solid = after

- **orientation:** contour normal
- **location:** contour position
- **output:** fixed probe p
- **decision:** finite labels

Counterexample: positive common scaling can keep the boundary while changing probabilities.



09 | LAB-D1-01 Launch

One-Neuron Boundary Workshop | 40 min

Commit: predict each isolated parameter effect.

Build: weighted sum, sigmoid, threshold, contour.

Observe: shapes, probe probabilities, old/new boundaries.

Return: one changed quantity + one invariant.

```
courseware/day-1/labs/LAB-D1-01-neuron-boundary.ipynb
```

10 | ACT-D1-02: Be the Neural Network

Start: $w_1=1$, $w_2=0$, $b=0$

Constraint: at most three one-parameter changes.

Before each change, record:

- which point(s) should change class;
- why the score sign should change;
- what remains fixed.

Score reasoning, not speed.

11 | LAB-D1-01 Debrief

Bring one contour and one probe trace.

1. Which parameter change affected location without orientation?
2. What changed even when sample accuracy did not?
3. Why is a workable setting not unique?
4. What would automate this search?

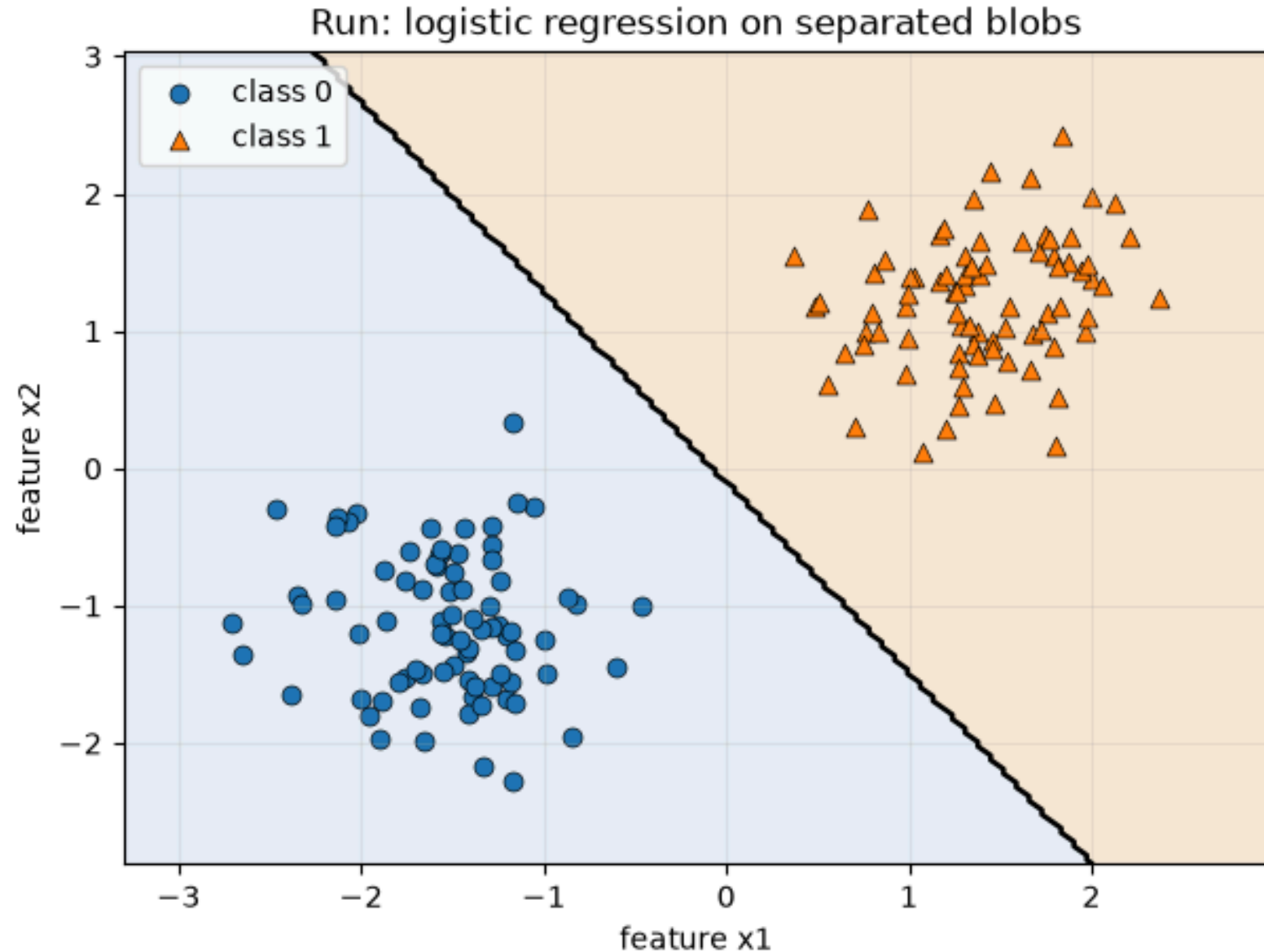
Bridge: an objective + a parameter-change rule.

12 | A Linear Boundary Succeeds

Prediction: Can one line separate these two clusters?

Reveal: decision region -> boundary -> accuracy

Ask: What did the score add beyond the picture?



13 | XOR: Draw One Separating Line

Corner	Target
low, low	0
low, high	1
high, low	1
high, high	0

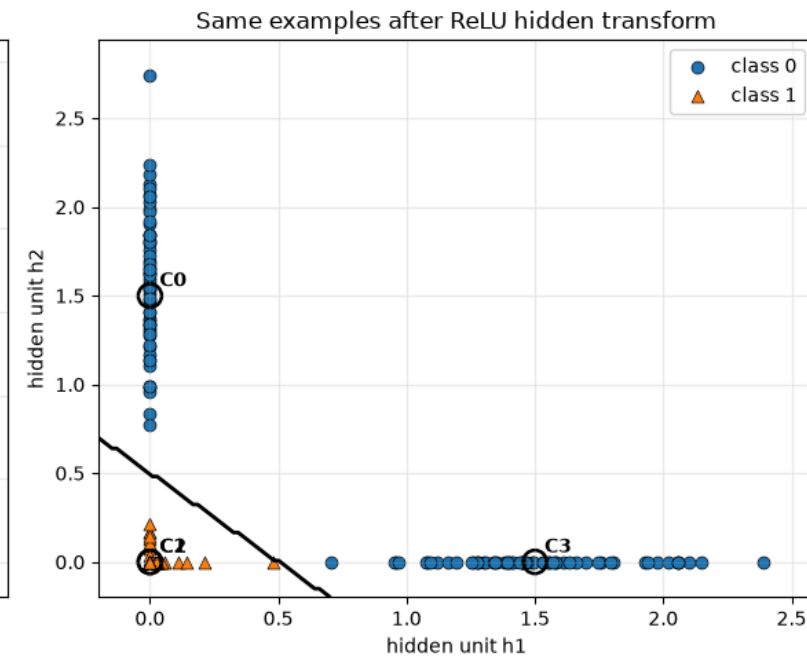
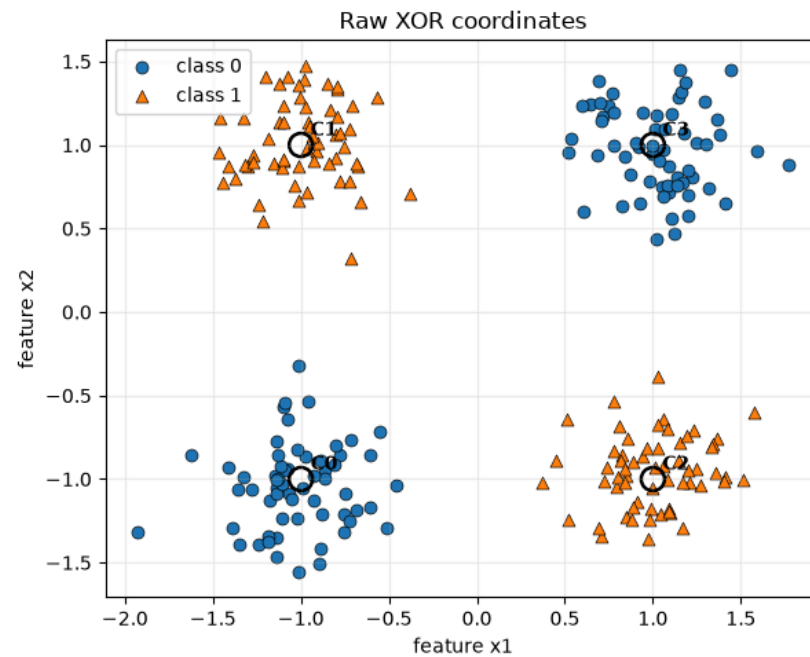
Commit: Can more epochs change the *kind* of boundary?

14 | XOR in Hidden Coordinates

Raw XOR -> hidden transform -> simple cut

Same points. New coordinates.

Caveat: hidden axes are not automatically human concepts.



15 | LAB-D1-02 Launch

The Linear Limit | 45 min

Rank expected performance:

1. logistic model on separable blobs;
2. logistic model on XOR;
3. fixed nonlinear hidden transform;
4. same stack with identity activations.

```
courseware/day-1/labs/LAB-D1-02-linear-limit.ipynb
```

16 | LAB-D1-02 Debrief

Use one raw/hidden-space comparison.

- Why did one neuron succeed first?
- Why did it fail on XOR?
- Did the output boundary become complex, or did the representation change?
- What evidence rejects "train longer"?

Key distinction: representation limit vs. optimization problem.

17 | Midday Checks

CHECK-D1-01

8 min: choose a first model and complete both boundary scenarios, including evidence probes.

CHECK-D1-02

5 min: make an unscored shape/parameter prediction for `5 -> 6 -> 3`.

2 min: submit D1-01 + a snapshot of the D1-02 commitment. Keep the working copy for revision after `LAB-D1-03`.

18 | A Layer Is a Shape Contract

X (B,d0)

-> **W1 (d0,d1) + b1 (d1,)**

A1 (B,d1)

-> **W2 (d1,d2) + b2 (d2,)**

logits (B,d2)

$$X_{(B,d_{in})} W_{(d_{in},d_{out})} + b_{(d_{out},)}$$

19 | ACT-D1-04: Shape Relay

Fill before code:

```
X | W1 | b1 | Z1 | A1 | W2 | b2 | logits | probabilities
```

Then:

1. count trainable parameters;
2. change only `B: 5 -> 32` ;
3. mark every shape that changes;
4. name one shape-correct semantic error.

20 | Parameters Are Not Examples

$$(4 \times 8 + 8) + (8 \times 3 + 3) = 67$$

Output task	Final representation
regression	numeric output
binary class	one logit -> sigmoid
3-way class	three logits -> softmax

Prompt: Which choice changes if B changes?

21 | More Linear Layers Are Still Linear

$$(XW_1 + b_1)W_2 + b_2 = X(W_1W_2) + (b_1W_2 + b_2)$$

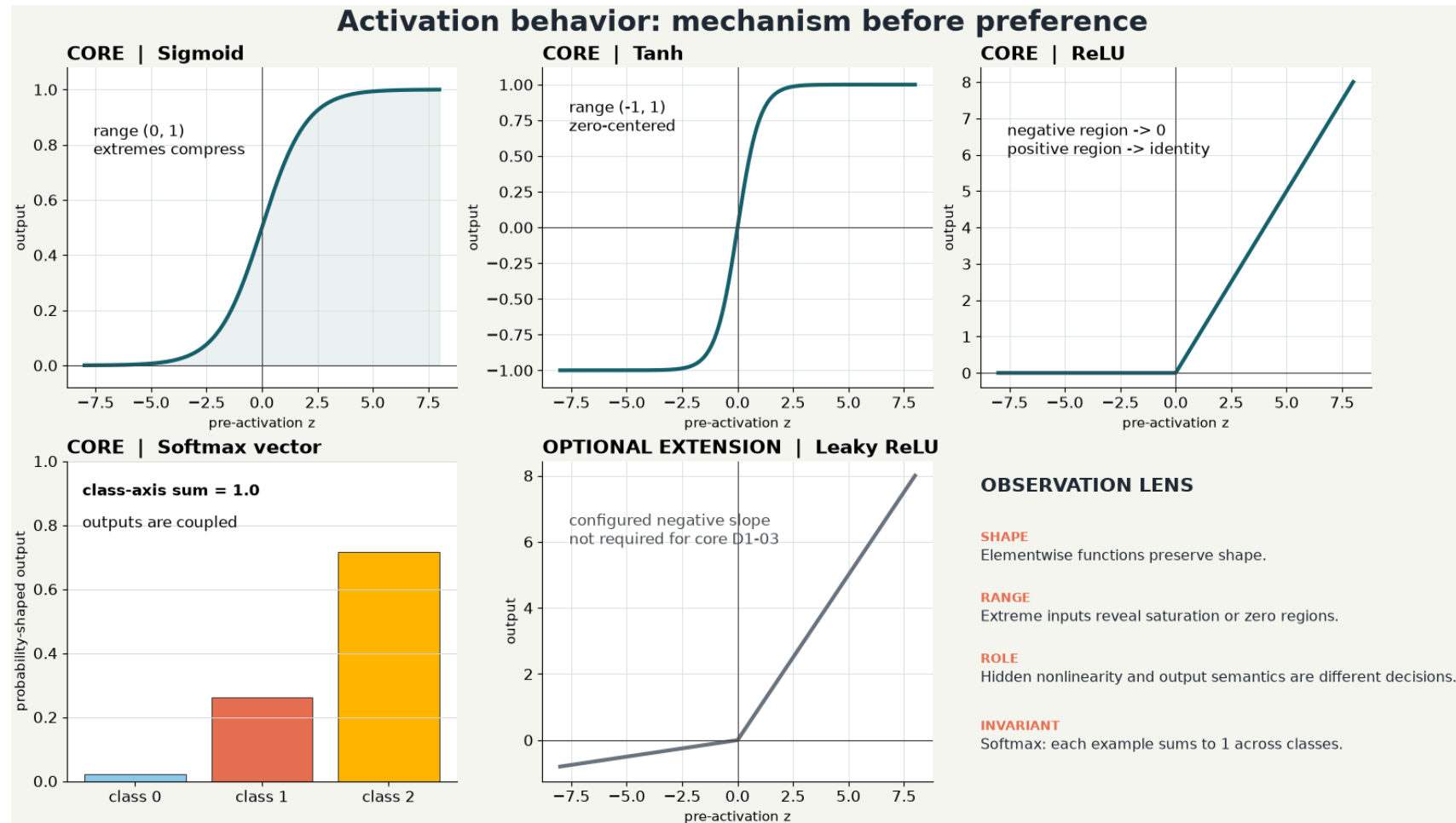
Identity activations: one composed affine map.

Question: What must change to alter the representational family?

22 | Activation Evidence

- sigmoid: $(0, 1)$, saturates
- tanh: $(-1, 1)$, saturates
- ReLU: negative $\rightarrow 0$
- softmax: coupled class vector

Optional: Leaky ReLU
keeps a negative slope.



23 | ACT-D1-03: Activation Card Sort

Match four core functions before labels appear:

function | curve | behavior | layer role | failure symptom

Then write an executable invariant for:

- sigmoid;
- ReLU;
- softmax.

Revise one match after the reveal and cite evidence.

Optional: add the Leaky ReLU card.

24 | LAB-D1-03 Launch

Shape and Activation Observatory | 40 min

Before each revealing cell, write:

expected shape | expected range | invariant

Watch for:

- saturation and zero regions;
- correct softmax row sums;
- wrong-axis output that still looks plausible.

```
courseware/day-1/labs/LAB-D1-03-shapes-activations.ipynb
```

25 | LAB-D1-03 Debrief

1. Why can wrong-axis softmax run?
2. Which invariant catches it?
3. What did extreme inputs do to sigmoid/tanh?
4. Does one zero ReLU output prove permanent failure?

Necessary: shape correctness

Not sufficient: semantic correctness

26 | Forward Propagation Is a Trace

$X \rightarrow Z1 \rightarrow A1 \rightarrow \text{logit } Z2 \rightarrow \text{sigmoid } P \rightarrow \text{class}$

At every arrow:

shape? range? invariant? fixed parameter?

27 | LAB-D1-04 Launch

Build a Network That Thinks Forward | 60 min

Before reveal: choose two likely failure probes.

Predict evidence in:

1. raw geometry;
2. probabilities;
3. decision region;
4. hidden representation.

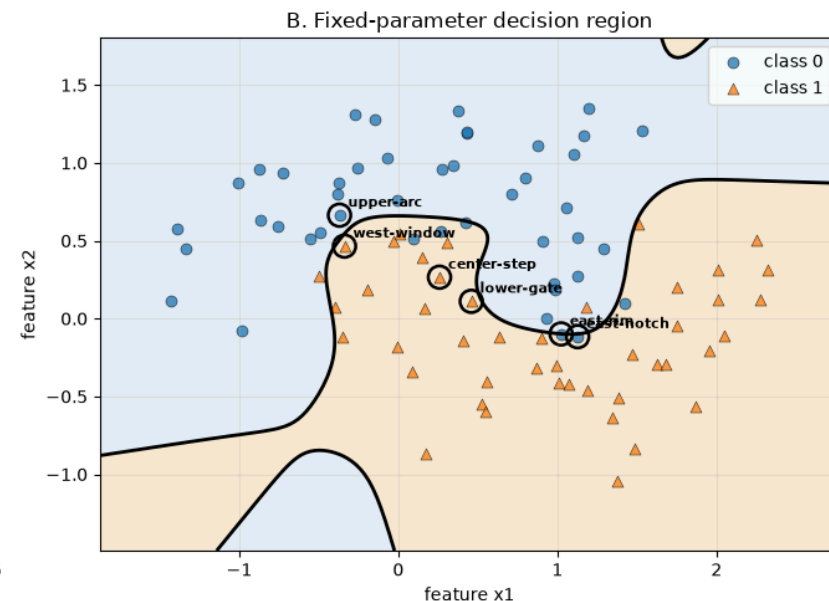
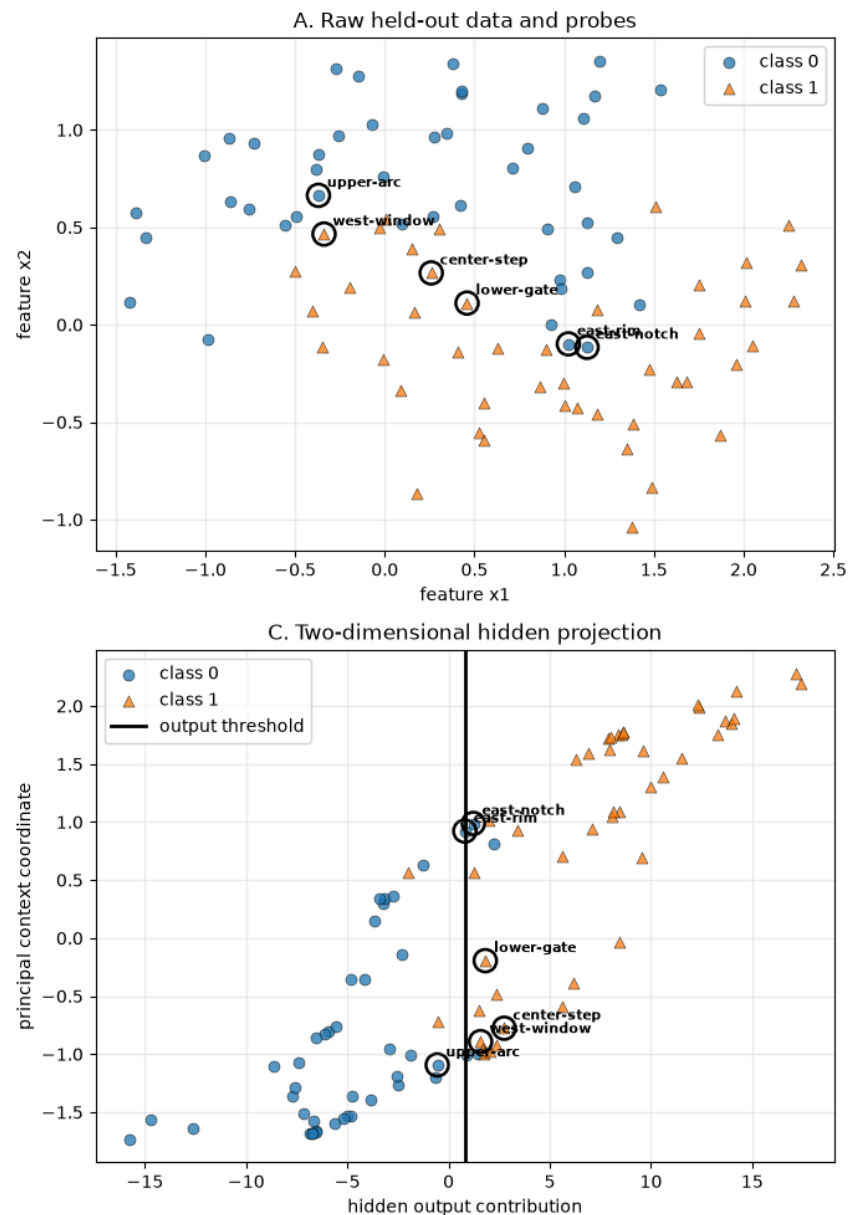
Use a **copy** for parameter perturbation.

```
courseware/day-1/labs/LAB-D1-04-forward-network-mystery.ipynb
```

28 | Forward Evidence Board

- **raw:** plausible failure?
- **region:** which side?
- **hidden:** easier separation?
- **trace:** values -> output?

Limit: evidence describes this trace, not full cause.



D. Six-probe forward trace

name	z1 range	a1 range	z2	p	result
west-window	[-10.00, 3.32]	[-1.00, 1.00]	0.79	0.687	correct
lower-gate	[-5.49, 3.53]	[-1.00, 1.00]	1.02	0.735	correct
upper-arc	[-10.23, 4.17]	[-1.00, 1.00]	-1.35	0.205	correct
center-step	[-6.64, 2.80]	[-1.00, 0.99]	1.95	0.876	correct
east-rim	[-6.17, 4.47]	[-1.00, 1.00]	0.01	0.502	wrong
east-notch	[-6.61, 4.54]	[-1.00, 1.00]	0.43	0.605	wrong

29 | LAB-D1-04 Debrief

Choose one correct and one incorrect probe.

For each:

1. cite an activation or logit;
2. cite the boundary/threshold evidence;
3. classify model error vs. implementation failure;
4. state what the trace cannot explain.

Replace "**the network knows**" with a parameter-and-activation statement.

30 | Exit Checks

CHECK-D1-03

Diagnose XOR and the "more identity layers" proposal.

CHECK-D1-04

Explain a wrong-but-complete forward trace and distinguish it from a shape failure.

Readiness gate: What must learning add?

Due before Day 2 at 09:00: five-minute consolidation: D1-03 item 5 + D1-04 items 2 and 5.

31 | Rebuild Your Day 1 Model

Complete aloud:

A neural network is a chain of _____ transformations.

A prediction is _____.

Hidden layers need _____ to escape one affine map.

A wrong output can still come from a _____ computation.

32 | Tomorrow: Make the Parameters Move

Day 1: input -> forward trace -> prediction

Day 2 adds:

prediction → error signal → sensitivity → parameter update → repeat

Exit prompt: Which Day 1 artifact should become the first half of that loop?